IT 5007: Software Engineering on Application Architecture
Lecture-10B

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Mobile Application Development

ReactNative

The BackStory

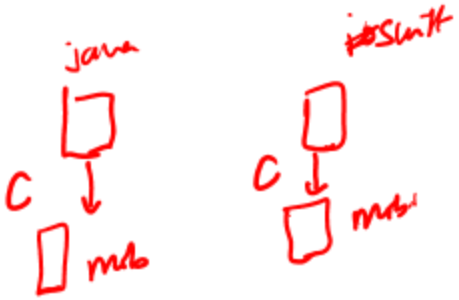
2015 @ Facebook

Conventional Development of FB App

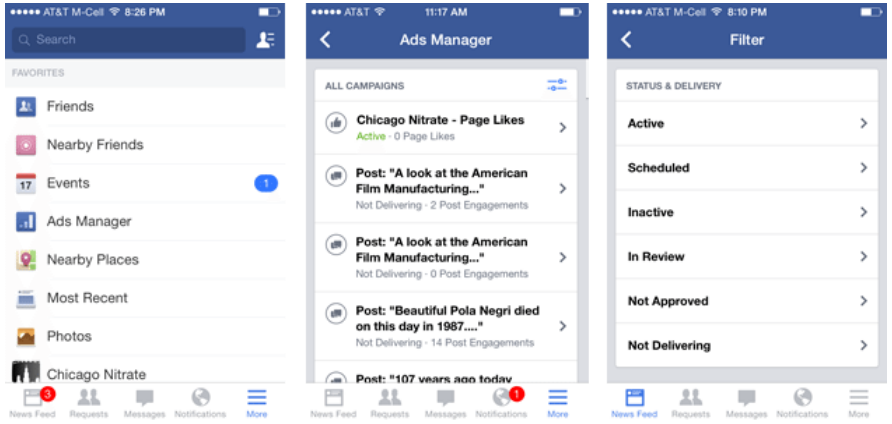
- Time taken to develop was high:
  - **Two Teams/Twice the time**: Android Team (Java), iOS team (Swift)
  - Sometimes the Apps **look very different** on Android and iOS.
  - **Iteration was slow**: Code-Compile-Deploy time was too high (Java/Swift)
  - Develop everything **from the Scratch**.

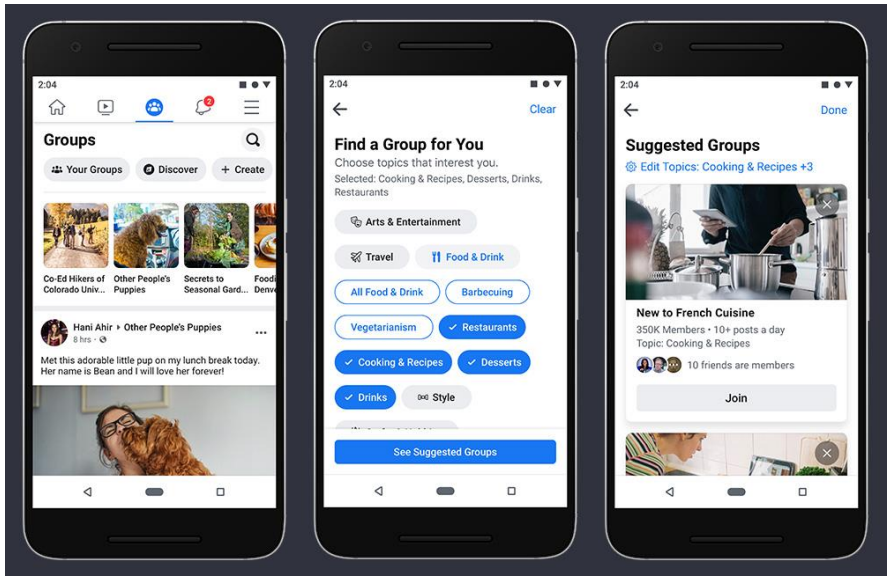
What could be the solution to make App dev platform agnostic?

Birth of ReactNative (2015)



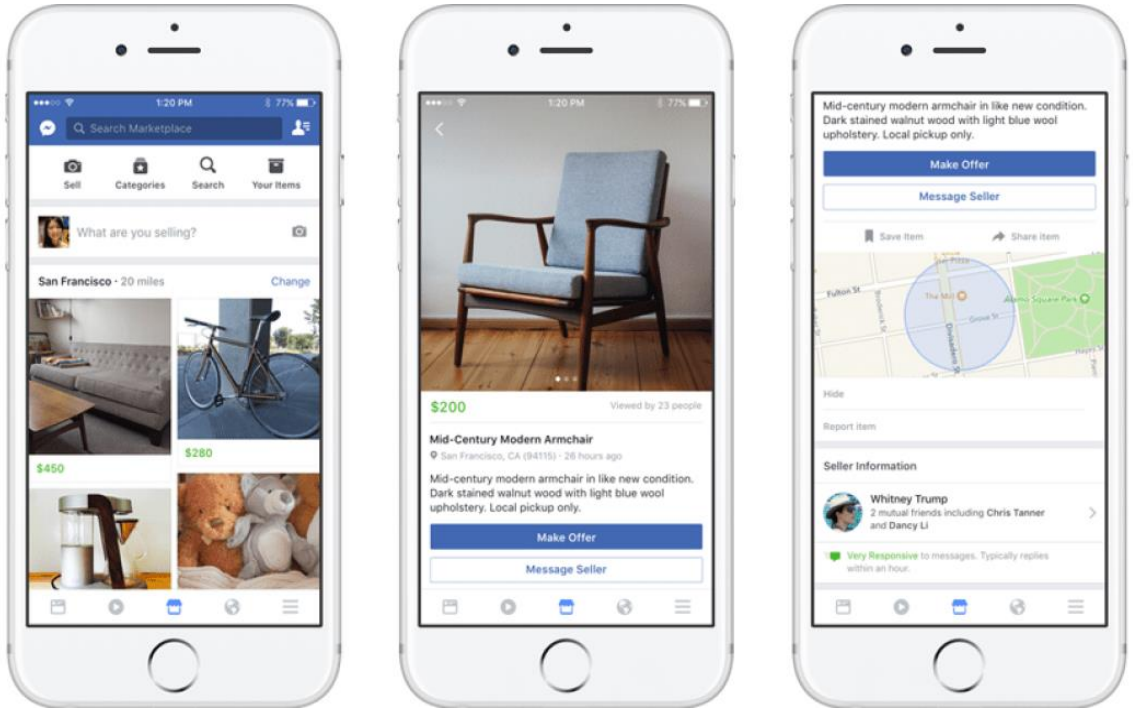
- Jordan Walke, Christopher Chedeau, Lin He, Ashwin Bharambe coded the **first Javascript code to render a Mobile UI element** (UILabel) in an iPhone (in 2 days).
- Coded a few other UI elements in the coming days.
- Organized a hackathon where participants need to use these JavaScript React Components to render iOS UI elements.
- Challenge: Developing JS equivalent for every Android/iOS UI element was tedious. What did they do?
- Solution: Open Source and ask community to contribute.
- First commercial use:





The Biggest User of ReactNative

Facebook MarketPlace



Interesting Resources

- https://www.techaheadcorp.com/blog/history-of-react-native/
- https://www.youtube.com/watch?v=NCAY0HIfrwc

ReactNative Concepts

Are we using the same git repository?

At the Heart: Components

Listing 2-6. App.jsx: JSX Part Separated Out from the HTML

```
const element = (
  <div title="Outer div">
    <h1>Hello World!</h1>
  </div>
);

ReactDOM.render(element, document.getElementById('contents'));
```

- Community-Contributed Components
- Custom Components written by user

RN vs. React

Similarities
- Component
- react state, hooks

- Components for Mobile UI
- Key Difference: These are components that interact with an underlying **mobile OS** for
  - Rendering Mobile UI components
  - To interface with device API

Location
Camera
File

- These components are called **Native Components**.
- How do they get rendered both on Android and iOS?

text label
view

```
import React from 'react';
import { Text, View } from 'react-native';

const HelloWorldApp = () => {
  return (
    <View
      style={{
        flex: 1,
        justifyContent: "center",
        alignItems: "center"
      }}>
      <Text>Hello, world!</Text>
    </View>
  )
}
export default HelloWorldApp;
```

UI

ReactNative: JSX to Native Transformation

REACT NATIVE UI COMPONENT	ANDROID VIEW	IOS VIEW	WEB ANALOG	DESCRIPTION
`<View>`	`<ViewGroup>`	`<UIView>`	A non-scrollling `<div>`	A container that supports layout with flexbox, style, some touch handling, and accessibility controls
`<Text>`	`<TextView>`	`<UITextView>`	`<p>`	Displays, styles, and nests strings of text and even handles touch events
`<Image>`	`<ImageView>`	`<UIImageView>`	`<img>`	Displays different types of images
`<ScrollView>`	`<ScrollView>`	`<UIScrollView>`	`<div>`	A generic scrolling container that can contain multiple components and views
`<TextInput>`	`<EditText>`	`<UITextField>`	`<input type="text">`	Allows the user to enter text

npm run android

CodePen Examples

ReactNative Counter

- https://codepen.io/pkarthik88/pen/WNJqOrp

React Counter

- https://codepen.io/pkarthik88/pen/eYrwRBK

Side note: Notice the difference in CodePen Environments

Types of Components

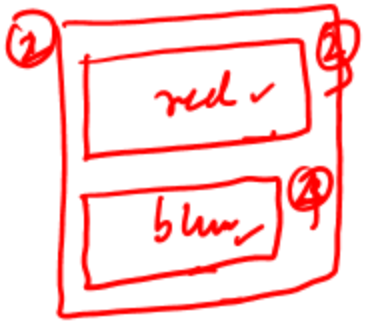
- **Core Component**: Readily available for use by the developer. (e.g., Text, TextInput, View, Button, etc.)
- **Community Contributed Components**: Install npm package and use the components readily.
- **User-written Custom Components**: Sometimes what you want to do only exists in Native Code. (e.g., FB team wanted to use the native (obj-C) based code for marketplace filters).

ReactNative: Behind the Scenes

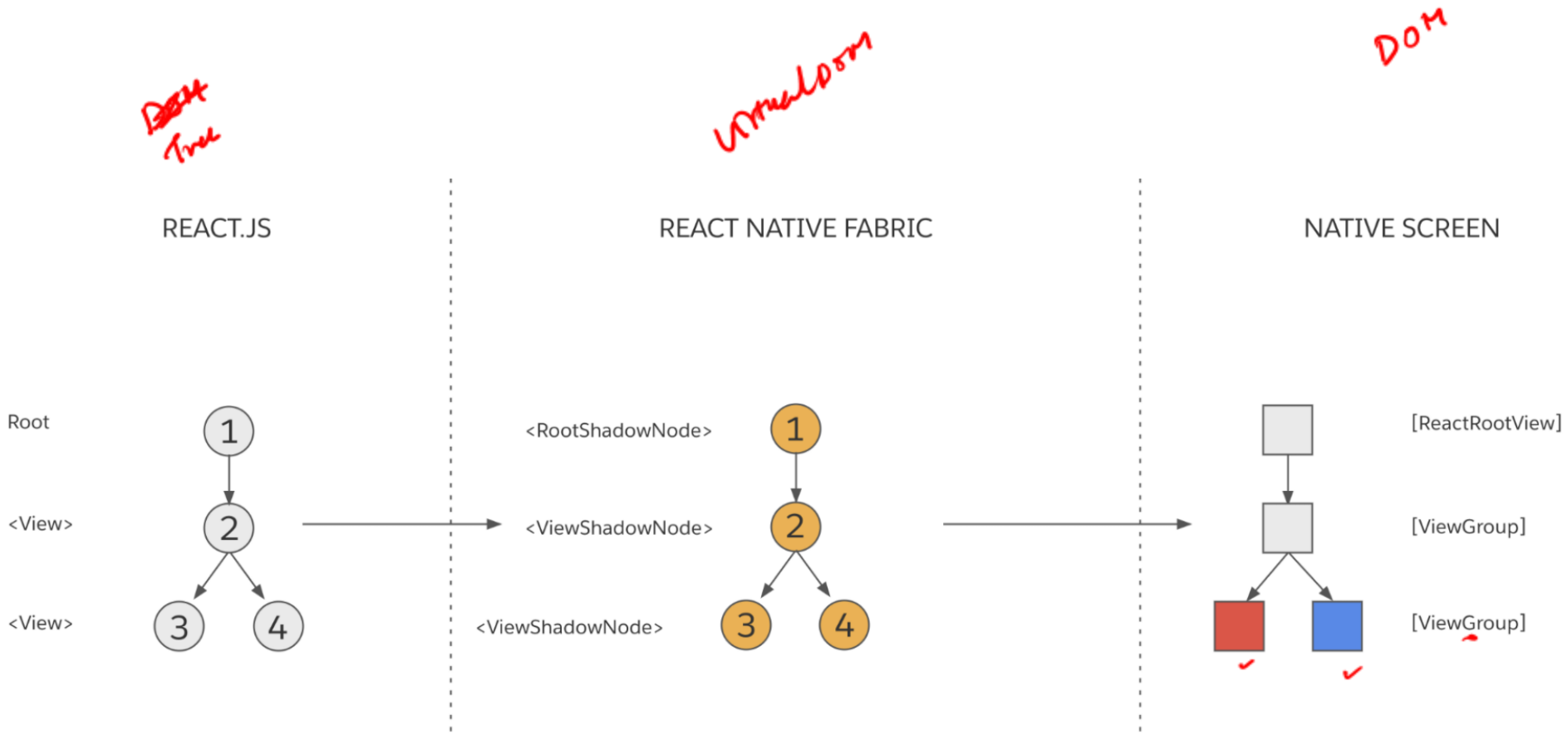
How does ReactNative Component get converted to Android Java Classes?

1. Writing code in React Native

```
function MyComponent() {
  return (
    <View>
      <View
        style={{ backgroundColor: 'red', height: 20, width: 20 }}
      />
      <View
        style={{ backgroundColor: 'blue', height: 20, width: 20 }}
      />
    </View>
  );
}
```



2. Rendering Process



React Element Tree

React Shadow Tree

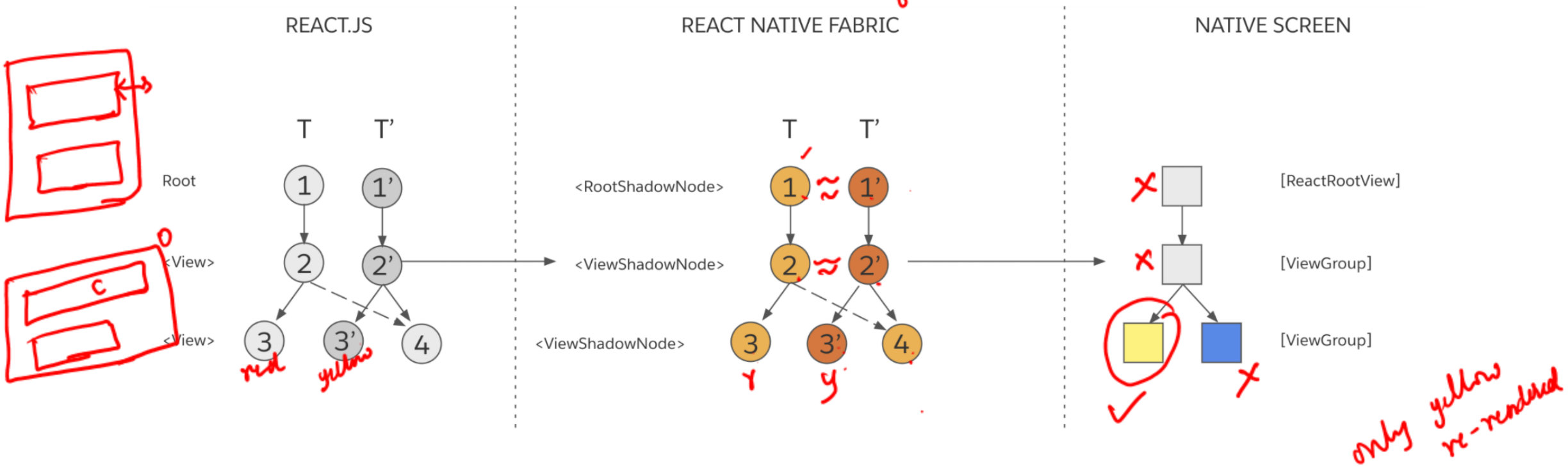
Host View on Host Platform

3. State Changes

state change

```
<View>
  <View
    style={{ backgroundColor: 'yellow', height: 20, width: 20 }}
  />
  <View
    style={{ backgroundColor: 'blue', height: 20, width: 20 }}
  />
</View>
```

4. Render on State Change



React Element Tree

React Shadow Tree

Host View on Host Platform

https://segmentfault.com/a/1190000041182092/en

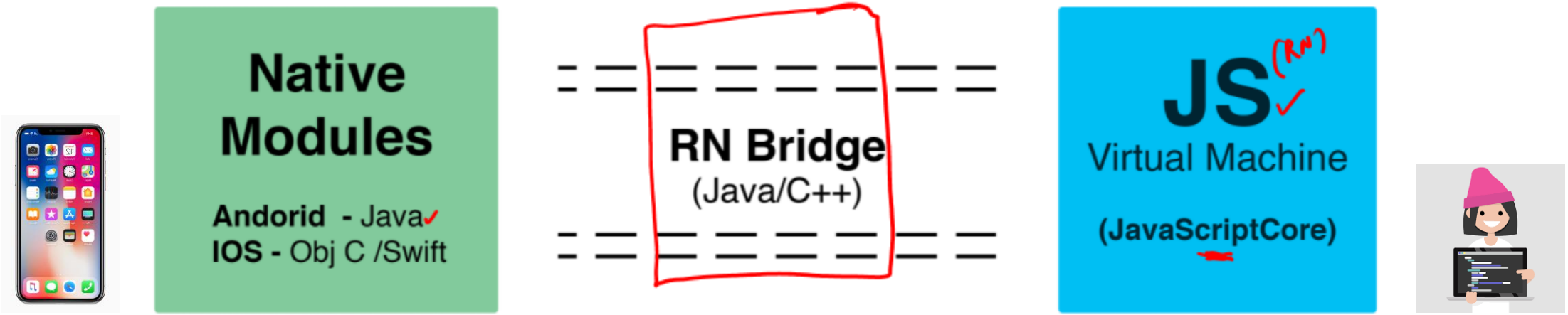
How does JS code execute then?

(v8)
browser – JS
BE (nodejs) – JS
Mobile (?) – JS

- Not a V8 engine.
- JavaScriptCore. Apple
- Is there a JavaScriptCore inside our Android device?
- Yes!

How can JS render something on Android Screen?

- React Native Bridge
- Details: https://www.reactnative.guide/3-react-native-internals/3.1-react-native-internals.html



ReactNative Architecture

JS
Virtual Machine
(JavaScriptCore)

1. <TextInput> sent to RN Bridge
2. Compute State Changes
3. <Text> sent to RN Bridge
4. Compute State Changes
5. <Text> sent to RN Bridge

RN Bridge <TextInput>

RN Bridge <EditText>

RN Bridge (Touch)

RN Bridge <Text>

RN Bridge <TextView>

RN Bridge (Scroll)

RN Bridge <Text>

RN Bridge <Textview>

JS Queue/Thread

Shadow Thread

Native/Main Queue/Thread

1. App Loaded with components
2. User Touches an image
3. App Reloads with components
4. User Scrolls view
5. App reloads with new components

Native Modules

Andorid - Java
IOS - Obj C /Swift